ALGEBRA

बीजगणित

SHEET-03

CLASS NOTES

CONCEPT OF SYMMETRY

* $a+b+c=0$

$a=b=c$

$3a=0$

$a=0$

Symmetry → power same

Symmetry → coefficients same

1. If $xy + yz + zx = 1$, then $\dfrac{1+y^2}{(x+y)(y+z)} = ?$

(a) 0 (b) 1
(c) 2 (d) 3

$x = y = z$

$\therefore \quad x^2 + x^2 + x^2 = 1$

$3x^2 = 1$

$x^2 = \frac{1}{3}$

$\dfrac{1+x^2}{2x \times 2x} = \dfrac{1+x^2}{4x^2}$

$= \dfrac{1+\frac{1}{3}}{4 \times \frac{1}{3}} = \dfrac{\frac{4}{3}}{\frac{4}{3}} = 1$

$x = y = z$

2. If $x^2 + y^2 + z^2 = xy + yz + zx$, then the value of $\left(\frac{17x^4 + 9y^4 + 16z^4}{8x^2y^2 + 6y^2z^2 + 10z^2x^2}\right)$ is:

SSC Phase-XII 24/06/2024 (Shift-02)

(a) 1.75 (b) 3.25

(c) 2.00 (d) 2.50

$\frac{42x^4}{24x^4} \quad \frac{7}{4} = 1.75$

3. If $a^2 + b^2 + c^2 = ab + bc + ca$ then $\frac{a+c}{b}$ is

(a) 0 (b) 2

(c) 1 (d) −1

put $a=b=c$

$a^2+a^2+a^2 \quad a^2+a^2+a^2$

$3a^2 = 3a^2$

$\frac{a+a}{a} = \frac{2a}{a}$

$x = y = z$

$\left(\frac{4x-3}{x}\right)+\left(\frac{4x-3}{x}\right)+\left(\frac{4x-3}{x}\right)=0$

$\Rightarrow 3x\left(\frac{4x-3}{x}\right)=0$

$\Rightarrow \frac{4x-3}{x}=0$

$\Rightarrow 4x-3=0$

$\Rightarrow x=\frac{3}{4}$

4. If $\frac{4x-3}{x}+\frac{4y-3}{y}+\frac{4z-3}{z}=0$ then $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is:

(a) 9 (b) 3

(c) 4 (d) 6

$= \frac{1}{x}+\frac{1}{x}+\frac{1}{x}$

$= \frac{3}{x} = \frac{3}{3}\times 4$

$\left(\frac{2+a}{a}\right) \times 3 = 4$

$\Rightarrow \quad 2+a = \frac{4a}{3}$

$\Rightarrow \quad 2 = \frac{4a}{3} - a$

$\Rightarrow \quad 2 = \frac{a}{3} \quad \Rightarrow 6 = a$

5. If $\frac{2+a}{a} + \frac{2+b}{b} + \frac{2+c}{c} = 4$ then the value of $\frac{(ab+bc+ca)}{abc}$ is

$= \frac{3a^2}{a^3} = \frac{3}{6} \frac{1}{2}$

(a) 2 (b) 1

(c) 0 (d) $\frac{1}{2}$

$a = b = c$

$a=b=c$

$\left(\frac{x+3a^2}{2a}\right)\times 3=0$

$\Rightarrow x+3a^2=0$

$\Rightarrow x=-3a^2$

6. If $\frac{x+a^2+2c^2}{b+c}+\frac{x+b^2+2a^2}{c+a}+\frac{x+c^2+2b^2}{a+b}=0$ $= 0$, find x

(a) $a^2+b^2+c^2$ $3a^2$ (b) $-(a^2+b^2+c^2)$ $-3a^2$

(c) $a^2+2b^2+c^2$ $4a^2$ (d) $-(a^2+2b^2+c^2)$ $-4a^2$

$\left(\frac{x-a^2}{2a^2}\right) \times 3 = 3 \times 1$

$\therefore \quad x - a^2 = 2a^2$

$x = 3a^2$

7. If $\frac{x-a^2}{b^2+c^2} + \frac{x-b^2}{c^2+a^2} + \frac{x-c^2}{b^2+a^2} = 3$, find the value of x.

a^2 (a) $a^2 + b^2 - c^2$ $3a^2$ (b) $a^2 + b^2 + c^2$

$-a^2$ (c) $a^2 - b^2 - c^2$ $2a^2$ (d) $a^2 + b^2$

$a = b = c$

a=b=c

8. If $bc + ca + ab = abc$ then

$$\frac{b+c}{bc(a-1)} + \frac{c+a}{ca(b-1)} + \frac{a+b}{ab(c-1)} = ?$$

(a) 0 (b) 1

(c) 2 (d) 3

$3a^2 = a^3$

$a = 3$

$\frac{2a}{a^2(a-1)} \times 3$

$= \frac{2}{3 \times 2} \times 3 \times 1$

$\frac{1}{1+a} \times 3$

$= \frac{1}{3} \times 3$

9. If $a^2 = b + c$, $b^2 = c + a$, $c^2 = a + b$, then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$

(a) 0 (b) 1

(c) 2 (d) 3

put $a=b=c$

$a^2 = 2a$

$a=2$

10. If a, b, c are non zero, $a+\frac{1}{b}=1$ & $b+\frac{1}{c}=1$ then (i) abc is (ii) $c+\frac{1}{a}$ is $=1$

(a) $-1, 1$ (b) $3, -1$

(c) $-3, 1$ (d) $1, 1$

$a=b=c$

$a+\frac{1}{a}=1$

$\Rightarrow a^2+1=a$

$\Rightarrow a^2+1-a=0$

$\Rightarrow (a+1)(a^2+1-a)=0$

$\therefore (a^3+1^3)=0$

$a^3=-1$

i) $a\times a\times a=a^3$
$=-1$

ii) $a+\frac{1}{a}=1$

Method-1 (Basic)

11. If $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$ and $a^z = (x + y + z)^x$, then $x + y + z = ?$ $(a \neq 0)$

(a) 0 (b) 1

(c) a^3 (d) a

$a^x = (x+y+z)^y$

$a^y = (x+y+z)^z$

$a^z = (x+y+z)^x$

$a^x \cdot a^y \cdot a^z = (x+y+z)^y \cdot (x+y+z)^z \cdot (x+y+z)^x$

$\Rightarrow a^{(x+y+z)} = (x+y+z)^{(x+y+z)}$

$\therefore a = x+y+z$

- $a^m \times a^n = a^{m+n}$
- $a^m = a^n \Rightarrow m = n$
- $m^a = n^a \Rightarrow m = n$

put $x=y=z$

$a^x = 3x^x$

$\therefore a=3x$

11. If $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$ and $a^z = (x + y + z)^x$, then $x + y + z = ?$ $(a \neq 0)$

(a) 0 (b) 1

(c) a^3 (d) a

$3x = a$

CONCEPT OF VALUE PUTTING

* $a+b=5$

1	4
2	3
−1	6

no. = 2 − 1
= (1)

$a+b=5$
$a-b=3$

$a=4$ $b=1$

2 − 2
= 0 ✗

Special Concept

\# $a+b+c=0$

$(-1, 1, 0)$

$(-2, 1, 1)$

12. If $a + b + c = 0$, then the value of $\dfrac{a^2+b^2+c^2}{a^2-bc}$

(a) 0 (b) 1

(c) 2 (d) 3

$a+b+c=0$

$-1, 1, 0$

$\therefore \dfrac{a^2+b^2+c^2}{a^2-bc} = \dfrac{1+1}{1-0} = \dfrac{2}{1}$

$a+b+c=0$

$-2 \quad 1 \quad 1$

$\dfrac{(4)+1+1}{4-1} = \dfrac{6}{3} = 2$

13. If $a^{-1} + b^{1} + c^{0} = 0$ then the value of $\frac{a^2 - bc}{b^2 - cd}$ is :

(a) 0 (b) 1

(c) 2 (d) 3

$\frac{a^2}{b^2} = \frac{1}{1}$

14. If $a + b + c = 0$ then the value of $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$ is:

(a) 2 (b) −2

(c) 0 (d) 4

$-1 \quad 1 \quad 0$

$\frac{1+1}{-1} = -2$

15. If $x + y = 1$, then what is the value of $x^3 + 3xy + y^3$?

SSC CGL MAINS (08/08/2022)

(a) 0
(b) 2
(c) 1
(d) −1

16. If $a + b = 6$ and $ab = 5$, then what is the value of $a^3 + b^3$?

SSC CGL MAINS (08/08/2022)

(a) 136 (b) 126

(c) 116 (d) 106

$a+b = 6$ $ab = 5$

$5 + 1$ 5×1

$a^3+b^3 = 125+1 = 126$

17. If $a^3 + b^3 = 217$ and $a + b = 7$, then the value of ab is:

SSC CPO 24/11/2020 (Shift- 02)

(a) –6 (b) –1

(c) 7 (d) 6

$a^3+b^3=217$ $a+b=7$

$216+1$ $6+1$

ab

$6\times1=6$

Q $x + \frac{4}{x} = 4$ $x = 2$

Q. $x + \frac{9}{x} = 6$ $x = 3$

Q. $x + \frac{16}{x} = 8$ $x = 4$

Q $x + \frac{25}{x} = 10$ $x = 5$

Format

$x + \frac{p^2}{x} = 2p$

$\therefore x = p$

18. If $r + \frac{64}{r} = 16$, then the value of $r^4 + \frac{1}{r^3}$ is.

SSC CHSL 25/05/2022 (Shift- 03)

(a) $4096\frac{1}{512}$ (b) 512

(c) $512\frac{1}{4096}$ (d) 4096

$r = 8$

$8^4 + \frac{1}{8^3}$

$= 4096 + \frac{1}{512}$

$= 4096\frac{1}{512}$

$2 + \frac{1}{3} = 2\frac{1}{3}$

$5 + \frac{1}{7} = 5\frac{1}{7}$

19. If $a - \frac{12}{a} = 1$, $a > 0$, find $a^2 + \frac{16}{a^2}$?

(a) 15 (b) 19

(c) 17 (d) 11

put $a = 4$

$4 - \frac{12}{4} = 1$

$16 + \frac{16}{16} 1$

$= 17$

20. If $(x+6y) = 8$, and $xy = 2$, where $x > 0$, what is the value of $(x^3 + 216\,y^3)$?

SSC CGL 11/04/2022 (Shift- 01)

(a) 288
(b) 224
(c) 368
(d) 476

$8 + 216$

$= 224$

21. If $a^2 + b^2 = 65$ and $ab = 8$, $a > b > 0$, then find the value of $a^2 - b^2$.

SSC CGL 21/04/2022 (Shift- 02)

(a) 65 **(b) 60**

(c) 72 **(d) 63**

$a^2+b^2=65$

$64+1=65$ ✓

$ab=8$

8×1 ✓

a^2-b^2

$=64-1$

$=63$

22. If $x^2 + 9y^2 = 40$ and $xy = 4$, where $x > 0$, $y > 0$, then what is the value of $(x^3 + 27y^3)$?

SSC PHASE IX 2022

(a) 224 (b) 416
(c) 440 (d) 800

$x^2+9y^2=40$
$4+36=40$ ✓

$xy=4$
2×2 ✓

$x>0$
$y>0$

x^3+27y^3
$= 8+27\times8$
$= 8+216$
$= 224$

23. If $xy = -6$ and $x^3 + y^3 = 19$ (x and y are integers), then what is the value of $\frac{1}{x^{-1}} + \frac{1}{y^{-1}}$?

$xy=-6$ } $x^3+y^3=19$

-2×3 ✓ } $-8+24=19$ ✓

$x+y = -2+3 = 1$

SSC CGL MAINS (08/08/2022)

(a) -2 (b) 2

(c) -1 (d) 1 ✓

$2a = -2b$

$a = -b$

put $b = 1$ $\therefore$ $a = -1$

$b = 2$ $\quad a = -2$

$b = 3$ $\quad a = -3$

24. If $2a + 2b + c = 0$, then find the value of

$$\frac{4a^2 + 4b^2 + 4c^2}{5c^2 - 8ab}.$$

SSC CPO 29/06/2024 (Shift-03)

(a) -1 (b) $-2\sqrt{3}$

(c) 1 (d) $2\sqrt{3}$

a=1 c=0

2+2b+0=0

⇒ 2b=-2

b=-1 ✓

24. If $2a + 2b + c = 0$, then find the value of

$$\frac{4a^2 + 4b^2 + 4c^2}{5c^2 - 8ab}.$$

SSC CPO 29/06/2024 (Shift-03)

(a) -1 (b) $-2\sqrt{3}$

(c) 1 ✓ (d) $2\sqrt{3}$

$$\therefore \frac{4\times1 + 4\times(-1)^2}{-8\times1\times(-1)} = \frac{4+4}{+8} = 1$$

$x+z=y$

$1+1=2$

25. If $x - y + z = 0$, then find the value of

$$\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy}.$$

SSC CGL 12/04/2022 (Shift- 03)

Ans $\frac{4}{2} - \frac{1}{4} - \frac{1}{4}$

$= \frac{8-1-1}{4} = \frac{6}{4}\frac{3}{2}$

(a) $\frac{3}{2}$ ✓

(b) $\frac{1}{2}$

(c) -6

(d) $-\frac{3}{2}$

26. If $a^{-2} + b^{1} + c^{1} = 0$, then what is the value of

$$\frac{(b+c)^2}{bc} + \frac{(c+a)^2}{ca} + \frac{(a+b)^2}{ab} ?$$

SSC CGL 18/08/2021 (Shift- 03)

(a) 1 (b) –3

(c) –1 (d) 3

$$\frac{4}{1} + \frac{(-1)^2}{-2} + \frac{(-1)^2}{-2}$$

$$= 4 - \frac{1}{2} - \frac{1}{2}$$

$$= 4 - 1$$

$$= 3$$

$a+b=c$

$1+1 \quad 2$

27. If a + b – c = 0, then what is the value of

$$\frac{(b-c)^2}{4bc}+\frac{(c-a)^2}{4ca}-\frac{(a+b)^2}{4ab}\ ?$$

SSC CHSL 06/08/2021 (Shift- 01)

(a) $-\frac{3}{4}$ ✓

(b) $\frac{3}{4}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

$$= \frac{(-1)^2}{8}+\frac{(1)^2}{8}-\frac{4}{4}$$

$$= \frac{1}{8}+\frac{1}{8}-1$$

$$= \frac{1+1-8}{8}=\frac{-6}{8}=-\frac{3}{4}$$

$-\frac{3}{4}$ | $\frac{1}{2}$

28. If $x^{-2} + y^{1} + z^{1} = 0$, then what will be the value of $\left(\frac{x^2}{yz}\right) + \left(\frac{y^2}{zx}\right) + \left(\frac{z^2}{xy}\right)$?

SSC CHSL 24/05/2022 (Shift- 01)

$\frac{4}{1} + \frac{1}{-2} + \frac{1}{-2}$

$= 4 - \frac{1}{2} - \frac{1}{2}$

$= 4 - 1 = 3$

$3 \times \frac{-3}{-2}$

(a) $\frac{3(x^2+y^2+z^2)}{xyz}$

(b) $x^2 + y^2 + z^2$

$4 + 1 + 1 = 6$

(c) $\frac{x^2y^2z^2}{x}$

(d) 3

$\frac{4 \times 1 \times 1}{-2} - 2$

29. If $a + b + c = 0$, then

$$\left(\frac{a+b}{c}+\frac{b+c}{a}+\frac{c+a}{b}\right)\left(\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}\right)$$ is :

(a) 8 (b) 9

(c) – 3 (d) 0

$-2 \quad 1 \quad 1$

$\left(\frac{-1}{1}+\frac{2}{-2}-\frac{1}{1}\right)\left(\frac{-2}{2}+\frac{1}{-1}+\frac{1}{-1}\right)$

$= (-1-1-1)(-1-1-1)$

$= (-3)(-3)$

$= 9$

a=b=c=1

30. If $x = \frac{a}{b} + \frac{b}{a}$, $y = \frac{b}{c} + \frac{c}{b}$, $z = \frac{c}{a} + \frac{a}{c}$, then what is the value of $xyz - x^2 - y^2 - z^2$ = ?

(a) –4 (b) –2

(c) –1 (d) –6

$x=2$, $y=2$ $z=2$

$\therefore xyz - x^2 - y^2 - z^2$

$= 8 - 4 - 4 - 4$

$= -4$

31. ***

If $a^3b = abc = 180$, a, b, c are positive integers, then the value of c is :

(a) 110 (b) 1

(c) 4 (d) 25

$a^3b = abc = 180$

$a^2 a^3 b = abc$

$a^2 = c$

$a = 1, c = 1$

$abc = 180$

$1 \times b \times 1 = 180$

$b = 180$

$a=1$

$x=2$ $y=0$

32. If $x = a+\frac{1}{a}$ and $y = a-\frac{1}{a}$ then $\sqrt{x^4+y^4-2x^2y^2}$ is equal to :

$=x^2$

$=4$

(a) $16a^2$ 16 8 (b) 8

(c) $\frac{8}{a^2}$ 8 4 (d) 4

33. If $x = 2 - p$, then $x^3+3x^2p+3xp^2+p^3$ is equal to :

(a) 12 (b) 6

(c) 8 (d) 4

$x + p = 2$

$(x+p)^3 = (2)^3$

$x^3+3x^2p+3xp^2+p^3 = 8$

34. Find the product of

$(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$.

SSC CGL 07/03/2020 (Shift- 03)

(a) $a^3 + b^3 + 8c^3 - 6abc$

(b) $a^3 + b^3 + 6c^3 - 6abc$

(c) $a^3 + b^3 + 8c^3 - 2abc$

(d) $a^3 + b^3 + 8c^3 - abc$

$x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$

$a^3+b^3+8c^3-6abc = (a+b+2c)(a^2+b^2+4c^2-ab-2bc-2ac)$

a=1 b=2 c=3

34. Find the product of

$(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$.

SSC CGL 07/03/2020 (Shift- 03)

(a) $a^3 + b^3 + 8c^3 - 6abc$ 1+8+216−36=189

(b) $a^3 + b^3 + 6c^3 - 6abc$

(c) $a^3 + b^3 + 8c^3 - 2abc$

(d) $a^3 + b^3 + 8c^3 - abc$

$(1+2+6)(1+4+36-2-12-6)$

$= 9 \times 21$

$= 189$

Basic

35. If $a + b - c = 5$ and $ab - bc - ac = 10$, then find the value of $a^2 + b^2 + c^2$.

SSC CGL 18/04/2022 (Shift- 01)

(a) 15 (b) 45

(c) 5 (d) 40

$$(a+b-c)^2 = a^2+b^2+c^2+2(ab-bc-ac)$$

$$\Rightarrow \quad 25 = R + 2\times 10$$

$$\Rightarrow \quad S = R$$

35. If $a + b = 5$ and $ab = 10$, then find the value of $a^2 + b^2$

SSC CGL 18/04/2022 (Shift- 01)

(a) 15
(b) 45
(c) 5
(d) 40

$(a+b)^2 = a^2+b^2+2ab$

$\Rightarrow 25 = k + 20$

$5 = k$

Basic

36. If $a + b + c = 11$ and $ab + bc + ca = 28$, then find the value of $a^3 + b^3 + c^3 - 3abc$.

SSC CGL 19/04/2022 (Shift- 03)

(a) 407 (b) 1639

(c) 2255 (d) 1093

$$a^3+b^3+c^3-3abc = (a+b+c)\left[(a+b+c)^2 - 3(ab+bc+ac)\right]$$

$$= 11\left[121-84\right]$$

$$= 11 \times 34$$

$$= 404$$

36. If $a + b = 11$ and $ab = 28$, then find the value of $a^3 + b^3$

SSC CGL 19/04/2022 (Shift- 03)

(a) 407 ✓
(b) 1639
(c) 2255
(d) 1093

$a^3 + b^3 = 343 + 64$

$= 404$

MISCELLANEOUS QUESTIONS

37. If $p = 11$, then the value of $p(p^2 + 3p + 3)$ is:

SSC CHSL 25/05/2022 (Shift- 03)

(a) 1629
(b) 1225
(c) 1727
(d) 1111

p^3+3p^2+3p

$=(p^3+3p^2+3p+1)-1$

$=(p+1)^3-1$

$=12^3-1$

$=1728-1$

$=1727$

38. If $p = 38$, then the value of $p(p^2 + 3p + 3)$ is _____.

H.W

SSC CHSL 06/06/2022 (Shift- 03)

(a) 39313
(b) 59319
(c) 39318
(d) 59318

Basic

39. If $x = 11$, then the value of $(x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1)$ is:

SSC CPO 28/06/2024 (Shift-03)

(a) 10 (b) 8

(c) 9 (d) 14

$$x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$$

$$= x^5 - 11x^4 - x^4 + 11x^3 + x^3 - 11x^2 - x^2 + 11x + x - 1$$

$$= x^5 - x^1 \cdot x^4 - x^4 + x^1 \cdot x^3 + x^3 - x^1 \cdot x^2 - x^2 + x^1 \cdot x^1 + x - 1$$

$$= x^5 - x^5 - x^4 + x^4 + x^3 - x^3 - x^2 + x^2 + x - 1$$

$$= x - 1$$

$$= 11 - 1$$

$$= 10$$

मात्र 1sec

39. If $x = 11$, then the value of $(x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1)$ is:

SSC CPO 28/06/2024 (Shift-03)

(a) 10
(b) 8
(c) 9
(d) 14

$= x - 1$
$= 11 - 1$
$= 10$

Q. $x = 9$

$$x^4 - 10x^3 + 10x^2 - 10x + 3$$

Soln $x^4 - 9x^3 - x^3 + 9x^2 + x^2 - 9x - x + 3$

$= x^4 - x \cdot x^3 - x^3 + x \cdot x^2 + x^2 - x \cdot x - x + 3$

$= \cancel{x^4} - \cancel{x^4} - \cancel{x^3} + \cancel{x^3} + \cancel{x^2} - \cancel{x^2} - x + 3$

$= -9 + 3$

$= -6$

$-x + 3$

$= -9 + 3$

$= -6$

Q. $x = 14$

$$x^7 - 18x^6 + 18x^5 - 18x^4 + 18x^3 - 18x^2 + 18x + 11$$

Soln

$$+x+11$$
$$= 17+11$$
$$= \underline{28}$$

Q. $x = 23$

$$x^7 - 24x^6 + 24x^5 - 24x^4 + 24x^3 - 24x^2 + 27x + 13$$

Soln

$$4x + 13$$

$$= 92 + 13$$

$$= 105$$

Special Concept

$$(a^4+b^4+a^2b^2) = (a^2+b^2+ab)(a^2+b^2-ab)$$

$$91 = 13 \times 7$$

* $a^2+b^2 = \frac{13+7}{2} = 10$
* $ab = \frac{13-7}{2} = 3$

$$a^2+b^2+ab = 13$$
$$+\ a^2+b^2-ab = 7$$
$$2(a^2+b^2) = 13+7$$
$$a^2+b^2 = \frac{13+7}{2}$$

40. If $a^4 + b^4 + a^2b^2 = 273$ and $a^2 + b^2 - ab = 21$, then one of the values of $\left(\frac{1}{a} + \frac{1}{b}\right)$ is:

SSC CGL 24/08/2021 (Shift- 01)

(a) $-\frac{9}{4}$

(b) $-\frac{3}{4}$

(c) $\frac{9}{8}$

(d) $\frac{3}{2}$

$(a^4+b^4+a^2b^2)=(a^2+b^2+ab)(a^2+b^2-ab)$

$273 = 13 \times 21$

$(a+b)^2 = a^2+b^2+2ab$

$\Rightarrow \quad " \quad = 17-8$

$\Rightarrow (a+b)^2 = 9$

$\therefore \boxed{a+b=3}$

$\frac{1}{a}+\frac{1}{b} = \frac{(a+b)}{ab} = \frac{3}{-4} = -\frac{3}{4}$

41. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and $x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of $(x - y)$ is:

SSC CGL MAINS 03/02/2022

$(x^4+y^4+x^2y^2)=(x^2+y^2+xy)(x^2+y^2-xy)$

$\frac{273}{16} = \frac{13}{4} \times \frac{21}{4}$

(a) $\frac{5}{2}$ (b) $\frac{3}{4}$

(c) $\frac{5}{4}$ (d) $\frac{3}{2}$

$(x-y)^2 = x^2+y^2-2xy$

$= \frac{\left(\frac{13}{4}+\frac{21}{4}\right)}{2} - 2\left(\frac{\frac{13}{4}-\frac{21}{4}}{2}\right)$

$= \frac{34}{8} - (-2)$

$(x-y)^2 = \frac{34+16}{8} = \frac{50}{8} \frac{25}{4}$

$(x-y) = \frac{5}{2}$

a=4 b=3 c=2

42. $ab(a-b)+bc(b-c)+ca(c-a)$ **is equal to :**

~~(a)~~ $(a+b)(b-c)(c-a)$ $7\times1\times(-2)$

~~(b)~~ $(a-b)(b+c)(c-a)$ $1\times5\times(-2)$

~~(c)~~ $(a-b)(b-c)(c-a)$ $1\times1\times(-2)$

(d) $(b-a)(b-c)(c-a)$ $(-1)\times1\times(-2)$

$=2$

$4\times3(4-3)+3\times2(3-2)+2\times4(2-4)$

$= 12+6-8\times2$

$= 2$

43. If $x^4 + x^2y^2 + y^4 = 273$ and $x^2 - xy + y^2 = 13$, then the value of xy is :

SSC CGL 05/03/2020 (Shift- 02)

(a) 4 (b) 8

(c) 10 (d) 6

$(x^4+x^2y^2+y^4) = (x^2+xy+y^2)(x^2-xy+y^2)$

$273 = 21 \times 13$

$xy = \frac{21-13}{2} = 4$

Q. $(x^4+x^2y^2+y^4) = 80$ $\quad (x^2-xy+y^2)=16$

$xy = ?$

Ans $(x^4+x^2y^2+y^4) = (x^2+xy+y^2)(x^2-xy+y^2)$

$80 = 5 \times 16$

$$xy = \frac{5-16}{2} = \frac{-11}{2} = -5.5$$

44. If $16a^4 + 36a^2b^2 + 81b^4 = 91$ and $4a^2 + 9b^2 - 6ab = 13$, then what is the value of $3ab$?

SSC CGL 04/03/2020 (Shift- 01)

(a) -3 (b) $\frac{3}{2}$

(c) 5 (d) $-\frac{3}{2}$

$(16a^4+36a^2b^2+81b^4) = (4a^2+6ab+9b^2)(4a^2-6ab+9b^2)$

91 7 13

$4a^2+6ab+9b^2 = 7$

$-\ 4a^2 - 6ab + 9b^2 = 13$

$12ab = -6$

$\div 4 \rightarrow 3ab = -\frac{6}{4} = -\frac{3}{2}$

44. If $16a^4 + 36a^2b^2 + 81b^4 = 91$ and $4a^2 + 9b^2 - 6ab = 13$, then what is the value of $3ab$?

SSC CGL 04/03/2020 (Shift- 01)

(a) -3 (b) $\frac{3}{2}$

(c) 5 (d) $-\frac{3}{2}$

$(16a^4+36a^2b^2+81b^4) = (4a^2+6ab+9b^2)(4a^2-6ab+9b^2)$

91 7 13

$4a^2+9b^2 = \frac{7+13}{2} = 10$

$6ab = \frac{7-13}{2} = -3$

$3ab = \frac{-3}{2}$

45. If $\frac{7(x^2+1)-17x}{7x} = 2$, then $\left(\frac{x+1}{\sqrt{x}}\right)^2 = ?$

(a) 6.45 (b) 6.53
(c) 6.86 (d) 6.31

$7x^2+7-14x = 14x$

$\Rightarrow \frac{7x^2}{7x} - \frac{31x}{7x} + \frac{7}{7x} = 0$

$\Rightarrow x + \frac{1}{x} = \frac{31}{7}$

$\frac{x^2+1+2x}{x}$

$= x + \frac{1}{x} + 2$

$= \frac{31}{7} + 2 = \frac{45}{7} = 6.45$

:

46. If $\frac{7(2\sqrt{2}-x)}{x} = 7 + \frac{14}{x^2}$ then

$\sqrt{(23x^{64} + 42x^{12} + 163)}$

(a) 12 (b) $12\sqrt{2}$

(c) $12\sqrt{3}$ (d) 24

$\frac{2\sqrt{2}-x}{x} = \frac{x^2+2}{x^2}$

$\Rightarrow 2\sqrt{2} - x = x + \frac{2}{x}$

$\Rightarrow 2\sqrt{2} = 2x + \frac{2}{x}$

$\therefore x + \frac{1}{x} = \sqrt{2}$

$\Rightarrow x^2 + \frac{1}{x^2} = 0$

$\Rightarrow x^4 + 1 = 0$

$\Rightarrow x^4 = -1$

$\sqrt{23(x^4)^{16} + 42(x^4)^3 + 163}$

$= \sqrt{23 \times 1 - 42 + 163}$

$= \sqrt{144}$

$= 12$

47. ***

If $8k^6 + 15k^3 - 2 = 0$, then the positive value of $\left(k + \frac{1}{k}\right)$ is:

SSC CGL 12/04/2022 (Shift- 03)

(a) $2\frac{1}{2}$ (b) $2\frac{1}{8}$

(c) $8\frac{1}{2}$ (d) $8\frac{1}{8}$

Let $k^3 = x^1$

$8x^2 + 15x - 2 = 0$

$x = -\frac{16}{8}, +\frac{1}{8}$

$x = \frac{1}{8}$

$k^3 = \frac{1}{8}$

$k = \frac{1}{2}$

$= \frac{1}{2} + 2$

$= 2 + \frac{1}{2}$

$= 2\frac{1}{2}$

$ab+ac=-bc$
$ab+bc=-ac$
$bc+ca=-ab$

48. If $(ab + bc + ca) = 0$, then what is the value of $\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}\right)$?

SSC CHSL 09/06/2022 (Shift- 01)

(a) 2 **(b) 0**

(c) 1 **(d) a + b + c**

$$\frac{1}{a^2+ab+ac} + \frac{1}{b^2+ab+bc} + \frac{1}{c^2+bc+ac}$$

$$= \frac{1}{a(a+b+c)} + \frac{1}{b(a+b+c)} + \frac{1}{c(a+b+c)}$$

$$= \frac{1}{(a+b+c)}\left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right] = \frac{1}{a+b+c}\left[\frac{bc+ac+ab}{abc}\right] = 0$$

~~49~~. If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the values of p and q are:

Quadratic

SSC CHSL 08/06/2022 (Shift- 03)

(a) $p = -\frac{3}{4}$ and $q = 4$

(b) $p = \frac{3}{4}$ and $q = 4$

(c) $p = \frac{3}{4}$ and $q = -4$

(d) $p = -\frac{3}{4}$ and $q = -4$

50. **What is the value of the following expression?**

let $x = 1$

$$2^2\left(\frac{x^a}{x^b}\right)^{(a+b)} \times 3^2\left(\frac{x^b}{x^c}\right)^{(b+c)} \times 6^{-2}\left(\frac{x^c}{x^a}\right)^{(a+c)}$$

SSC CHSL 08/06/2022 (Shift- 02)

(a) 1 (b) 0

(c) 4 (d) 9

$4 \times 1 \times 9 \times 1 \times \frac{1}{36} \times 1$

$a^2 - b^2 = (a-b)(a+b)$

$a^4 - b^4 = (a^2-b^2)(a^2+b^2)$

$a^6 - b^6 = (a^3-b^3)(a^3+b^3)$

$a^8 - b^8 = (a^4-b^4)(a^4+b^4)$

51. **Simplify the following expression.**

$$\frac{5(a^6-b^6)^3+5(b^6-c^6)^3+5(c^6-a^6)^3}{2(a^3-b^3)^3+2(b^3-c^3)^3+2(c^3-a^3)^3}$$

SSC CGL 24/08/2021 (Shift- 02)

(a) $\frac{5}{2}(a^3+b^3)(b^3+c^3)(c^3+a^3)$

(b) $\frac{5}{2}(a^3+b^3)(b^3-c^3)(c^3-a^3)$

(c) $\frac{5}{2}(a^3-b^3)(b^3+c^3)(c^3+a^3)$

(d) $\frac{5}{2}(a^3-b^3)(b^3-c^3)(c^3+a^3)$

$$=\frac{5}{2}\left[\frac{(a^6-b^6)^3+(b^6-c^6)^3+(c^6-a^6)^3}{(a^3-b^3)^3+(b^3-c^3)^3+(c^3-a^3)^3}\right]$$

$$=\frac{5}{2}\frac{3(a^6-b^6)(b^6-c^6)(c^6-a^6)}{3(a^3-b^3)(b^3-c^3)(c^3-a^3)}$$

$$=\frac{5}{2}(a^3+b^3)(b^3+c^3)(c^3+a^3)$$

$$\frac{(a^6-b^6)}{(a^3-b^3)}=\frac{(a^3-b^3)(a^3+b^3)}{(a^3-b^3)}$$

Q. $$\frac{(a^4-b^4)^3+(b^4-c^4)^3+(c^4-a^4)^3}{(a^2-b^2)^3+(b^2-c^2)^3+(c^2-a^2)^3}$$

Ans $$\frac{3\times(a^4-b^4)(b^4-c^4)(c^4-a^4)}{3\times(a^2-b^2)(b^2-c^2)(c^2-a^2)}$$

$(a^2-b^2)(a^2+b^2)$

Trick: $(a^2+b^2)(b^2+c^2)(c^2+a^2)$

0. $$\frac{(a^8-b^8)^3+(b^8-c^8)^3+(c^8-a^8)^3}{(a^4-b^4)^3+(b^4-c^4)^3+(c^4-a^4)^3} = (a^4+b^4)(b^4+c^4)(c^4+a^4)$$

0. $$\frac{(a^8-b^8)^3+(b^8-c^8)^3+(c^8-a^8)^3}{(a^4+b^4)^3+(b^4+c^4)^3+(c^4+a^4)^3} = (a^4-b^4)(b^4-c^4)(c^4-a^4)$$

51. Simplify the following expression.

$$\frac{5(a^6-b^6)^3+5(b^6-c^6)^3+5(c^6-a^6)^3}{2(a^3-b^3)^3+2(b^3-c^3)^3+2(c^3-a^3)^3}$$

SSC CGL 24/08/2021 (Shift- 02)

(a) $\frac{5}{2}(a^3+b^3)(b^3+c^3)(c^3+a^3)$

(b) $\frac{5}{2}(a^3+b^3)(b^3-c^3)(c^3-a^3)$

(c) $\frac{5}{2}(a^3-b^3)(b^3+c^3)(c^3+a^3)$

(d) $\frac{5}{2}(a^3-b^3)(b^3-c^3)(c^3+a^3)$

* Number System → (20-28)
* HCF LCM → (5-6)
* Algebra → 20-22
* Quadratic Eq^{n} → 2
* Trigonometry → 20-22
* Height & Distance → (5-6)
* Geometry → (45)
* Mensuration 2D → 15
* " 3D → 12-15

Live R_1 R_2 off
4th, 5th, 6th, 7th 8th
X X X

52. If $x^3 + 4x = 8$, find $x^7 + 64\,x^2$?

(a) 32 (b) 64

(c) 128 (d) 256

$x^3+4x=8$

$\Rightarrow (x^3)^2=(8-4x)^2$

$\Rightarrow x^6 = 64+16x^2-64x$

$\Rightarrow x^7 = 16x^3-64x^2+64x$

$\Rightarrow x^7+64x^2 = 16x^3+64x$

$= 16(x^3+4x)$

$= 16\times 8$

$= 128$

53. If $x^2 - 5x + 25 = 0$ then $x^5 - 5x^4 - 3100$?

(a) 25 (b) $25\sqrt{2}$

(c) $25\sqrt{3}$ (d) None

$(x+5)(x^2-5x+25)=0$

$\Rightarrow (x^3+5^3)=0$

$x^3=-125$

$x^3(x^2-5x)-3100$

$=(-125)(-25)-3100$

$=+3125-3100$

$=25$

$(x^3+5^3)=(x+5)(x^2+25-5x)$

$$\frac{(x+5)(x+6)x(x-1)+9}{x(x-1)} = 0$$

$\Rightarrow (x^2+5x)(x^2-x+6x-6)+9=0$

$\Rightarrow (x^2+5x)(x^2+5x-6)+9=0$

$t \quad\quad t$

$\therefore t(t-6)+9=0$

$\Rightarrow t^2-6t+9=0$

$\Rightarrow (t-3)^2=0$

$t=3$

$x^2+5x=3$

$3(x^2+5x)-1$

$=3\times3-1$

$=8$

***

54. If $\frac{(x+5)(x+6)}{1} + \frac{9}{x(x-1)} = 0$,

Find $3x^2 + 15x - 1$?

(a) 4 **(b) 6**

(c) 8 **(d) 10**

Note:

$$(ax+by) = c$$
$$(ay-bx) = d$$

$$(a^2+b^2)(x^2+y^2) = c^2+d^2$$

55. If $ax + by = 8$, $ay - bx = 6$, $(a^2 + b^2) + (x^2 + y^2) = 29$.

Find $\frac{a^2 + b^2}{(x^2 + y^2)}$; $a^2 + b^2 > x^2 + y^2$?

(a) $\frac{25}{4}$ (b) $\frac{27}{4}$

(c) $\frac{23}{4}$ (d) $\frac{29}{4}$

$ax+by = 8$

$ay-bx = 6$

$(a^2+b^2)(x^2+y^2) = 8^2+6^2$

$mx\ n = 100$

$m+n = 29$

25 4

$\therefore \frac{a^2+b^2}{x^2+y^2} = \frac{m}{n} = \frac{25}{4}$

56. If $\frac{a^2+b^2+c^2-1024}{ab-bc-ca} = -2$ and $a+b = 5c$, where $c > 0$, then the value of c is _____.

(a) 8 (b) 4

(c) 12 (d) 5

$a^2+b^2+c^2-1024 = -2(ab-bc-ac)$

$\Rightarrow a^2+b^2+c^2+2(ab-bc-ac) = 1024$

$\Rightarrow (a+b-c)^2 = 1024$

$\Rightarrow a+b-c = 32$

$\Rightarrow 4c = 32$

$c = 8$

57. If $a^3 - b^3 = 602$ and $a - b = 2$, then find the value of $(a^2 + b^2)$.

IB ACIO GRADE II 17/01/2024 (Shift-01)

(a) 156 (b) 240

(c) 202 (d) 260

$a^3 - b^3 = 602$ $a - b = 2$

$1331 - 729$ $11 - 9$

$\therefore a^2 + b^2$

$= 121 + 81$

$= 202$

58. If $x = 7 + 7^{\frac{2}{3}} + 7^{\frac{1}{3}}$, then which of the options below is correct.

IB ACIO GRADE II 17/01/2024 (Shift-02)

(a) $x^3 + 21x^2 - 126x - 252 = 0$

(b) $x^3 + 21x^2 - 126x + 252 = 0$

(c) $x^3 - 21x^2 - 126x + 252 = 0$

(d) $x^3 - 21x^2 + 126x - 252 = 0$ ✓

$(x-7) = 7^{\frac{2}{3}} + 7^{\frac{1}{3}}$

$\Rightarrow (x-7)^3 = (7^{2/3} + 7^{1/3})^3$

$\Rightarrow x^3 - 21x^2 + 147x - 343 = \left(7^{\frac{2}{3}}\right)^3 + \left(7^{\frac{1}{3}}\right)^3 + 3 \times 7^{\frac{2}{3}} \times 7^{\frac{1}{3}} \left(7^{\frac{2}{3}} + 7^{\frac{1}{3}}\right)$

$\Rightarrow x^3 - 21x^2 + 147x - 343 = 49 + 7 + 3 \times 7 \times (x-7)$

$= 56 + 21x - 147$

$\Rightarrow x^3 - 21x^2 + 126x - 252 = 0$

$\rightarrow a^3 + b^3 + 3ab(a+b)$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

59. Select the option that is true regarding the following labelled Assertion (A) and Reason (R).

Assertion (A): The value of

$$\frac{\left[(3.9)^3+9\times1.3\times4.29+11.7\times(1.1)^2+1.331\right]}{\left[1.23\times1.23+0.77(0.77+0.6\times4.1)\right]}$$

is 31.25.

$$\frac{(3.9+1.1)^3}{(1.23+0.77)^2}$$

$$= \frac{125}{4} = 31.25$$

$(3.9)^3+3\times(3.9)^2\times1.1+3\times3.9\times(1.1)^2+(1.1)^3$

$(1.23)^2+(0.77)^2+2\times1.23\times0.77$

Reason (R): Using $(a+b)^3 = a^3+b^3+3ab(a+b)$ and $(x+y)^2 = x^2+y^2+2xy$

IB ACIO GRADE II 18/01/2024 (Shift-01)

(a) A is true and R is false.

(b) Both A and R are true but R is not a correct explanation of A.

(c) Both A and R are true and R is a correct explanation of A.

(d) A is false and R is true.

put $x=y=z$

$3x=1$

$x=\frac{1}{3}$

60. If x, y and z are positive numbers and $x+y+z=1$, then the least value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is:

SSC CGL 09/09/2024 (Shift-03)

(a) 7
(b) 5
(c) 9
(d) 11

$\frac{3}{x}=\frac{3}{\frac{1}{3}}=9$